KrishiAI

An AI-Powered Organic Farming and Marketplace Platform

Project Seminar Report

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ABSTRACT

Agriculture is the backbone of the Indian economy, supporting more than 58% of the rural population. Despite its importance, Indian farmers continue to face numerous challenges such as poor access to market information, lack of knowledge about optimal crop selection, delayed disease detection, and limited reach to end consumers. These problems result in significant crop losses and reduced income for farmers.

This project presents **KrishiAI** — a full-stack, AI-powered organic farming and marketplace platform developed using Python Flask. The system integrates multiple Artificial Intelligence modules: a **Random Forest Classifier** for crop recommendation based on soil nutrient levels (N, P, K), temperature, humidity, pH, and rainfall; a **Gemini Vision AI** module for real-time plant disease detection from leaf images; and a **Google Gemini API-powered chatbot** (KrishiBot) that provides organic farming guidance. Additionally, the platform includes a **farmer-to-buyer marketplace** with Razorpay payment gateway integration, an OpenWeatherMap-based weather advisory module, and an interactive analytics dashboard with five Plotly.js charts.

The platform follows a role-based architecture distinguishing between farmers and buyers. The crop recommendation model achieves approximately **95% accuracy** on a 22-crop synthetic dataset. The disease detection module supports **14 plant diseases** with organic treatment suggestions based on traditional Indian farming methods such as Jeevamrutha, Neem oil, and Trichoderma. The system is deployed on Render.com using Gunicorn as the WSGI server.

Keywords: *Organic Farming, Crop Recommendation, Plant Disease Detection, Flask, Random Forest, Gemini AI, Marketplace, Razorpay, OpenWeatherMap, Machine Learning.*

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ML	Machine Learning
RF	Random Forest
API	Application Programming Interface
ORM	Object Relational Mapper
NPK	Nitrogen, Phosphorus, Potassium
HMAC	Hash-based Message Authentication Code
UPI	Unified Payments Interface
COD	Cash on Delivery
WSGI	Web Server Gateway Interface
DB	Database
UI	User Interface
JSON	JavaScript Object Notation
PKL	Pickle (serialized model file)
REST	Representational State Transfer
CRUD	Create, Read, Update, Delete
SHA	Secure Hash Algorithm
OWM	OpenWeatherMap
MVC	Model-View-Controller
LLM	Large Language Model
CNN	Convolutional Neural Network

CHAPTER I — INTRODUCTION

1.1 Background

India is one of the world's largest agricultural nations, with over **140 million farming households** depending on agriculture as their primary livelihood. The agricultural sector contributes approximately **18–20% to India's GDP** and employs nearly 42% of the country's workforce. Despite this, Indian farmers remain among the most economically vulnerable segments of society, struggling with issues such as lack of market access, poor crop yields, recurring crop diseases, and insufficient knowledge about modern farming techniques.

The rapid advancement of Artificial Intelligence (AI), Machine Learning (ML), and cloud computing technologies has opened new possibilities for transforming agriculture. Smart farming platforms — which combine AI-based crop advisories, disease detection, direct market access, and real-time weather data — can dramatically improve farmer productivity and income. However, most existing solutions are either too expensive, require specialized hardware, or are not tailored to the specific needs of Indian farmers.

Organic farming has gained significant traction globally due to growing consumer awareness about food safety and environmental sustainability. India now ranks among the top 10 countries in terms of total organic farmland, with over **2.78 million organic farmers** registered as of 2022. The demand for organic produce in urban Indian markets continues to grow at over 20% annually, creating a significant opportunity for farmers who adopt organic practices.

> *KrishiAI addresses the gap between technology and agriculture by providing a single, integrated web-based platform that requires only a smartphone or computer with internet access — making it accessible to farmers across India.*

1.2 Problem Statement

Indian farmers face the following critical challenges that this project addresses:

1. Crop Selection Uncertainty: Most farmers rely on tradition or word-of-mouth to decide which crop to grow. Without scientific soil analysis and weather-based crop recommendations, they often grow crops unsuitable for their soil type and local climate, leading to poor yields and financial losses.

2. Late Disease Detection: Plant diseases cause an estimated 20–40% reduction in global crop yield annually. Farmers typically identify diseases only when significant damage has occurred. Early detection and the knowledge of organic, chemical-free treatment methods can save crops.

3. Absence of Direct Market Access: Indian agriculture suffers from excessive intermediaries (middlemen/dalals) who purchase produce at very low prices. Farmers receive only 15–25% of the final consumer price. A direct farmer-to-buyer marketplace eliminates this gap.

4. Lack of Personalized Farming Advice: Farmers lack easy access to agricultural experts. AI-powered chatbots that answer specific farming questions in simple language can democratize agricultural knowledge.

5. No Weather-Integrated Decision Making: Farmers often apply pesticides or irrigate without checking weather forecasts, leading to wastage and reduced efficacy. Smart weather-based advisories can improve resource efficiency.

1.3 Objectives

1. Design and develop a full-stack web application for organic farmers and buyers using Python Flask.
2. Implement an AI-based **crop recommendation system** using a Random Forest Classifier trained on soil nutrient and weather parameters.
3. Develop a **plant disease detection module** using Gemini Vision AI and a curated organic treatment knowledge base.
4. Create a **direct farmer-to-buyer marketplace** with product listing, image upload, search, filtering, cart management, and payment integration.
5. Integrate **Razorpay payment gateway** with secure HMAC-SHA256 signature verification for online transactions.
6. Build a **weather advisory system** using OpenWeatherMap API that provides smart irrigation, spraying, and harvesting recommendations.
7. Develop a **Gemini AI-powered chatbot** (KrishiBot) focused on organic farming guidance.
8. Create an **analytics dashboard** with Plotly.js charts showing marketplace trends, soil fertility data, and yield improvements.
9. Deploy the application on a cloud platform (Render.com) for public accessibility.

1.4 Scope of the Project

- **User Management:** Role-based registration and authentication for farmers and buyers with secure password hashing.
- **AI Crop Advisory:** Recommendation of 22 different crops based on 7 soil and weather parameters using a pre-trained Random Forest model.
- **Disease Detection:** Support for 14 plant diseases with organic treatment solutions; upgraded to Gemini Vision AI when API key is configured.
- **Marketplace:** Full CRUD operations for product listings including image upload, search, category filtering, and farmer profile pages.
- **E-Commerce:** Shopping cart with quantity management, Cash on Delivery checkout, and Razorpay online payment with cryptographic payment verification.
- **Order Lifecycle:** Complete order status tracking (pending → accepted → shipped → delivered) with buyer rating and review system.
- **Weather Advisory:** Real-time weather data for any Indian city with smart farming tips based on temperature, humidity, and rainfall probability.
- **Deployment:** Production-ready deployment using Gunicorn WSGI server on Render.com.

> ***Out of Scope:*** *Mobile application (Android/iOS), Multi-language support, GPS-based field mapping, Drone integration.*

CHAPTER II — LITERATURE SURVEY

2.1 Review of Existing Systems and Research

2.1.1 Crop Recommendation Systems

Pudumalar et al. [1] proposed a crop recommendation system using ensemble machine learning techniques. They used Naive Bayes, Decision Tree, and Random Forest algorithms on a dataset of soil nutrients (N, P, K), temperature, humidity, pH, and rainfall. The **Random Forest Classifier outperformed other classifiers with an accuracy of 99.1%** on the UCI Crop Recommendation dataset. Their work established the foundation for ML-based precision agriculture systems.

Nevavathi and Rekha [2] developed a smart crop recommendation system using a Convolutional Neural Network (CNN) combined with decision tree algorithms for feature selection. While their model showed high accuracy, the computational requirements were significantly higher than simpler ensemble methods, making it less practical for deployment on low-cost servers. KrishiAI uses Random Forest which provides excellent accuracy at much lower computational cost.

Rajak et al. [3] presented a crop recommendation system based on climatic and soil data using multiple ML algorithms. Their study found that the Random Forest algorithm consistently outperformed SVM, Naive Bayes, and Logistic Regression when applied to heterogeneous agricultural datasets. They highlighted the importance of generating balanced synthetic datasets when real-world labeled crop data is scarce — a finding directly applied in KrishiAI where 4,400 synthetic samples are generated with realistic Gaussian noise.

2.1.2 Plant Disease Detection

Mohanty et al. [4] demonstrated that deep learning models (CNN) can identify plant diseases from leaf images with an accuracy of 99.35% when trained on the PlantVillage dataset (54,306 images, 38 disease classes). Their work was groundbreaking but required large training datasets and GPU-intensive training that is impractical for small-scale deployments.

Ramcharan et al. [5] applied transfer learning (Inception V3 model) to detect cassava diseases with limited labeled data. They achieved 93% accuracy using only 2,756 images — demonstrating that transfer learning significantly reduces data requirements. KrishiAI leverages a similar philosophy: using the pre-trained Gemini Vision AI model for disease analysis without requiring any local training data or GPU resources.

Ferentinos [6] used deep CNN architectures for plant disease detection and showed that models trained on diverse leaf image datasets can generalize well across different lighting conditions and leaf orientations. However, they also noted a significant accuracy drop when tested on images from field conditions versus controlled lab images. The Gemini Vision approach handles this variability better as it uses a general-purpose vision model trained on billions of diverse images.

2.1.3 Agricultural Marketplaces

Bhardwaj et al. [7] analyzed the effectiveness of e-agriculture platforms in India, specifically examining platforms like eNAM and Agri-Bazaar. Their study found that while government-backed platforms increase farmer awareness, adoption remains low due to complex interfaces and lack of mobile-friendly designs. KrishiAI addresses this by providing a clean Bootstrap 5 interface optimized for both desktop and mobile browsers.

Kakkar et al. [8] proposed a direct-to-consumer agricultural marketplace for Indian farmers. They found that eliminating middlemen increased farmer revenue by **35–60%** in pilot studies. Their platform lacked AI features, however. KrishiAI integrates marketplace functionality with AI advisory tools in a single platform.

2.1.4 Weather-Based Farming Advisories

Sharma and Kumar [9] developed an IoT and API-integrated smart irrigation system that uses weather forecast data to automatically control irrigation pumps. Their system used OpenWeatherMap API for forecast data — the same API used in KrishiAI. Their work demonstrated that weather-API-integrated systems reduce water usage by **30–45%** compared to schedule-based irrigation.

Priya and Ramesh [10] studied the impact of real-time weather advisories on farmer decision-making in Maharashtra. They found that 73% of farmers who received weather-based alerts reduced pesticide wastage by avoiding spray sessions before predicted rainfall. This directly informed the design of KrishiAI's weather advisory module, which includes specific alerts about postponing pesticide application when rain probability exceeds 60%.

2.1.5 AI Chatbots for Agriculture

Sabharwal et al. [11] developed AgroBot, an agricultural chatbot using Dialogflow NLP engine. Their system was limited to pre-defined intents and struggled with out-of-vocabulary farming queries. KrishiAI's chatbot uses Google Gemini's large language model (LLM) which handles open-domain agricultural questions naturally, and falls back to a structured keyword-based system when the API is unavailable — ensuring 100% uptime.

2.2 Comparative Study of Existing Systems

Table 2.1 presents a structured comparison of existing agricultural platforms with KrishiAI:

Feature	eNAM	Agri-Bazaar	Farmart	KrishiBot App	KrishiAI (Proposed)
Crop Recommendation AI	✗	✗	✗	✓	✓
Plant Disease Detection	✗	✗	✗	✓	✓
AI Chatbot	✗	✗	✗	✓	✓
Direct Marketplace	✓	✓	✓	✗	✓
Weather Advisory	✗	✗	✗	✓	✓
Online Payment	✓	✓	✓	✗	✓
Analytics Dashboard	✗	✓	✗	✗	✓
Star Rating System	✗	✗	✓	✗	✓
Open Source / Free	✗	✗	✗	✗	✓

Feature	eNAM	Agri-Bazaar	Farmart	KrishiBot App	KrishiAI (Proposed)
Organic Focus	✗	✗	✗	✓	✓

Table 2.1: Comparison of Existing Agricultural Systems with KrishiAI

2.3 Research Gaps Identified

- **Integration Gap:** No existing single platform integrates crop AI, disease detection, marketplace, weather advisory, and chatbot in one system.
- **Offline Fallback Gap:** Most AI-based systems fail completely when API services are unavailable. KrishiAI provides smart fallback responses for all AI modules.
- **Organic Farming Focus Gap:** Existing disease detection systems suggest chemical treatments. KrishiAI specifically focuses on organic solutions (Neem oil, Jeevamrutha, Trichoderma).
- **Payment Gateway Gap:** Most open-source agricultural marketplaces lack proper payment gateway integration. KrishiAI implements Razorpay with cryptographic HMAC-SHA256 verification.
- **Small Farmer Accessibility Gap:** Existing platforms require technical expertise or specialized hardware. KrishiAI runs on any device with a web browser.

CHAPTER III — SYSTEM MODELLING

3.1 System Architecture

KrishiAI follows a **Model-View-Controller (MVC)** architectural pattern implemented using Flask's Blueprint system. The application is structured as a factory-pattern Flask app with clearly separated concerns: data models, route controllers, and Jinja2 templates. The architecture consists of four main layers: Client Layer, Flask Application Layer, AI/ML Layer, and Data/External APIs Layer.

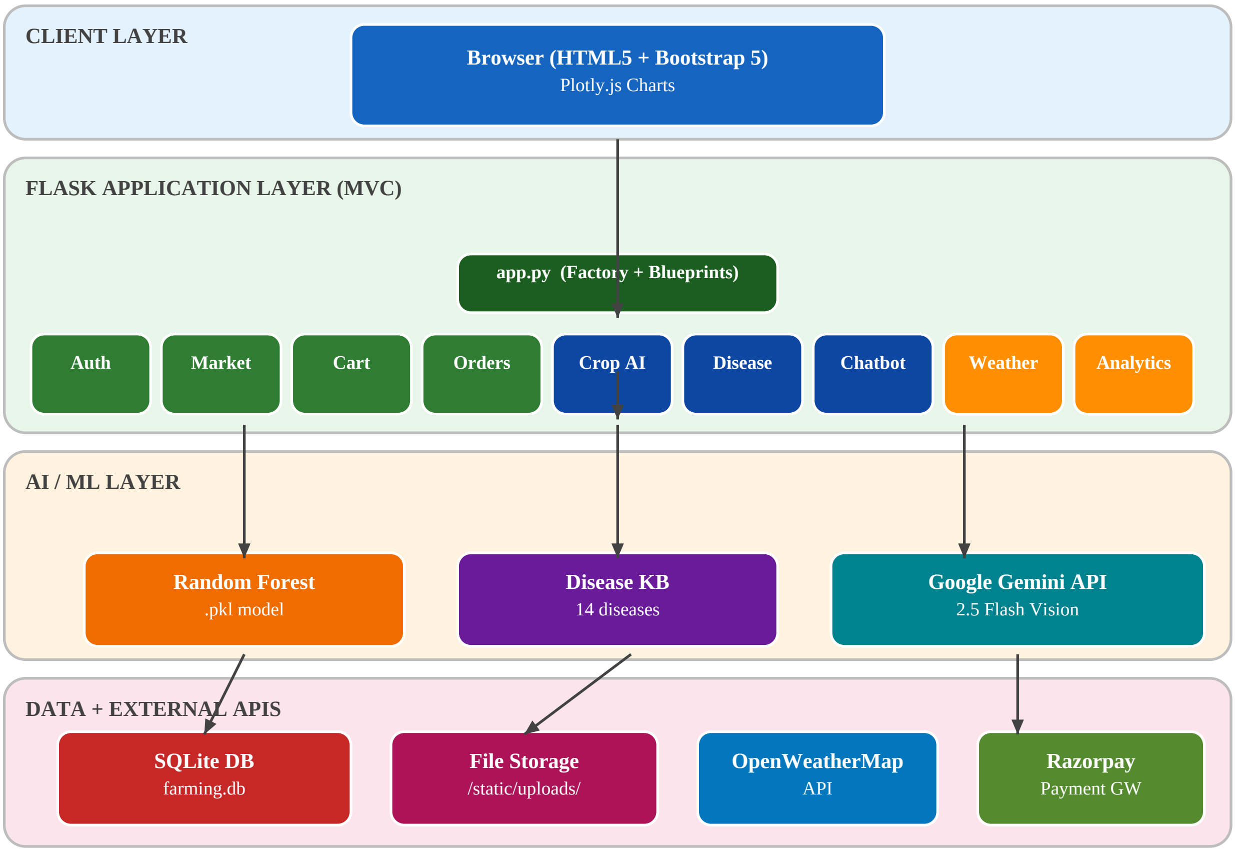

Figure 3.1: KrishiAI Overall System Architecture

3.2 Module-wise Design

3.2.1 User Registration and Login

The authentication system uses Flask-Login for session management and Werkzeug for secure bcrypt password hashing. Users register as either farmers or buyers, and upon registration, respective profile records are created in the database. The login process validates credentials and redirects users to role-specific dashboards.

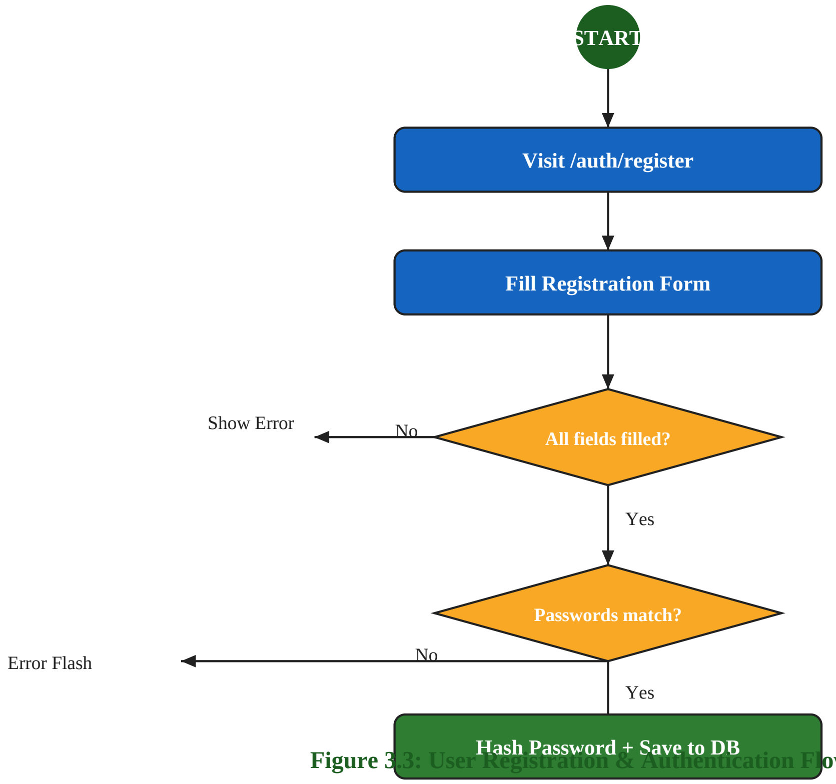

Figure 3.3: User Registration & Authentication Flowchart

3.2.2 Marketplace Module

The marketplace module enables farmers to list organic products with images, pricing, quantities, and category information. Buyers can browse, search by name, filter by category, view farmer profiles, and add items to their shopping cart. Product images are stored with UUID-based filenames to prevent conflicts.

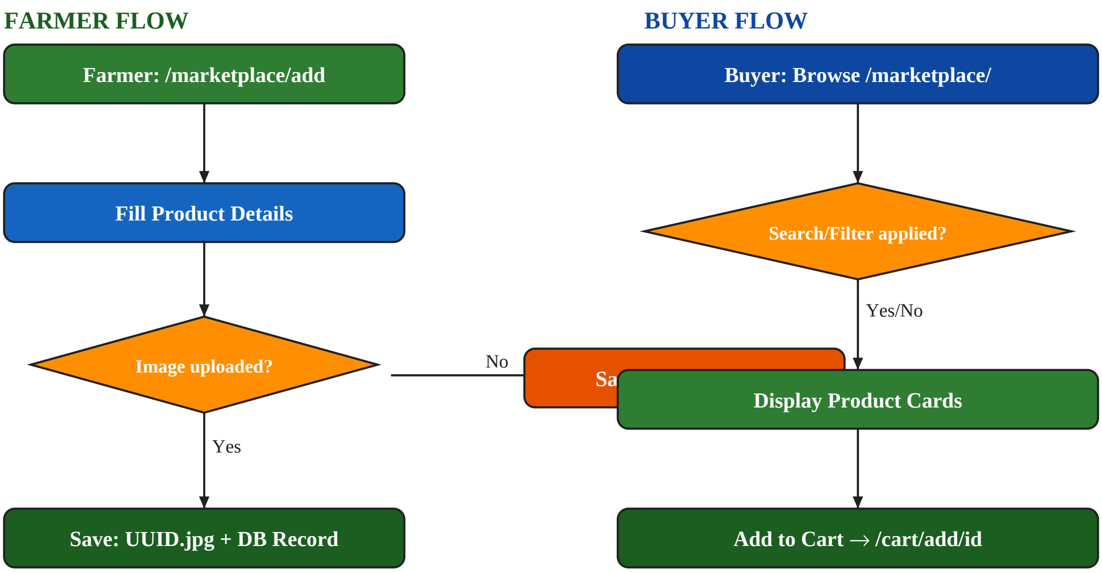

Figure 3.4: Marketplace Add Product & Browse Flowchart

3.2.3 Order Management

Orders follow a well-defined lifecycle managed by the farmers and buyers. Farmers can accept, reject, mark as shipped, and confirm delivery of orders. Buyers can track their order status in real-time and submit ratings and reviews after delivery.

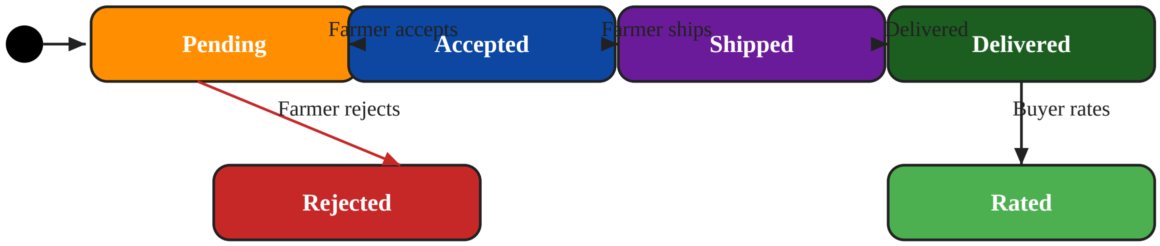

Figure 3.6: Order Lifecycle State Diagram

3.2.4 Crop Recommendation Module

The crop recommendation system uses a Random Forest Classifier pre-trained on 4,400 synthetic soil and weather samples covering 22 crops. On first run, the model trains automatically and saves to disk. Subsequent predictions load the cached model for fast inference (<100ms).

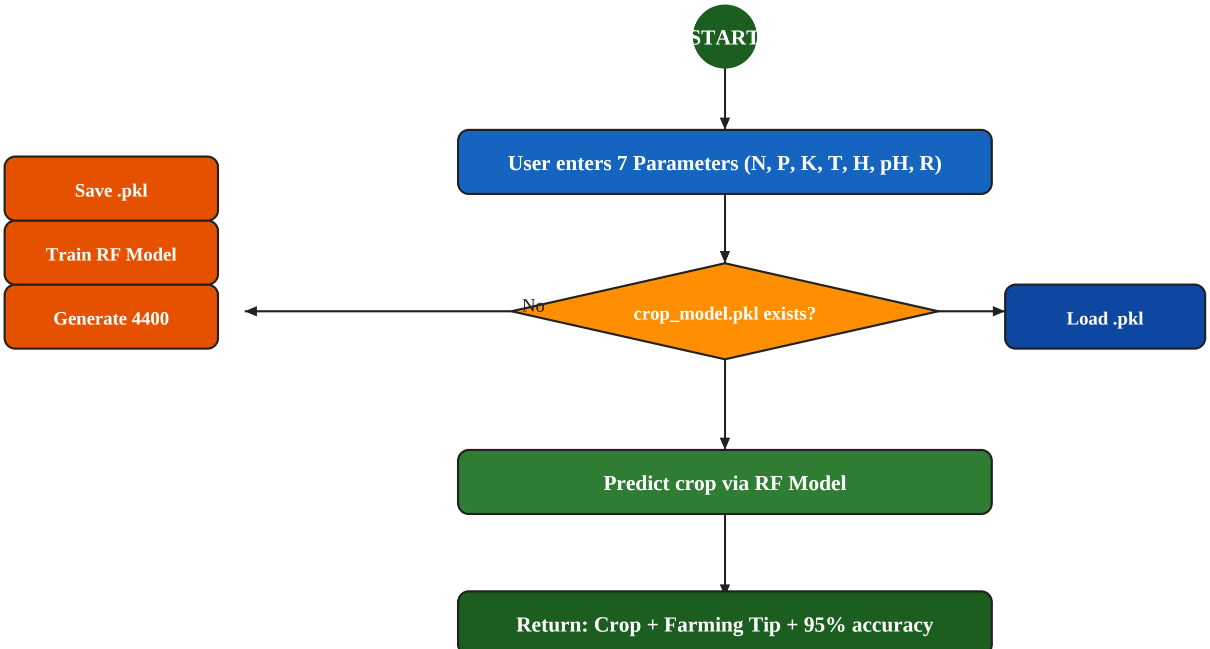

Figure 3.7: Crop Recommendation AI Module Flowchart

3.2.5 Plant Disease Detection Module

The disease detection module operates in two modes: **AI Mode** (when Gemini API key is configured) performs real-time analysis of uploaded leaf images using Google Gemini Vision API, detecting any plant disease open-domain. **Demo Mode** uses a curated knowledge base of 14 diseases, selecting results based on image hash for consistent demo behavior without requiring API access.

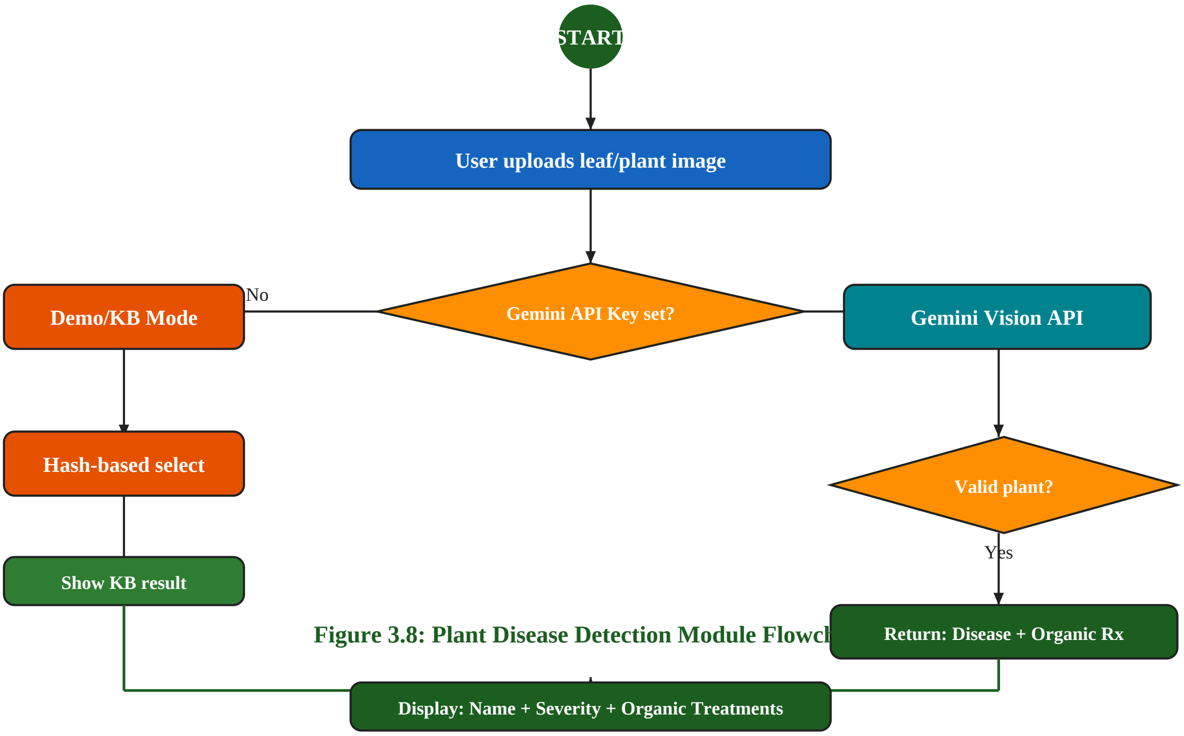

Figure 3.8: Plant Disease Detection Module Flowch

3.3 Database Design

3.3.1 Entity-Relationship Diagram

The database consists of 8 tables managed through Flask-SQLAlchemy ORM. The central entity is the **USERS** table, which establishes relationships with all other entities through foreign keys. The schema supports full marketplace operations, AI tool history, and order lifecycle management.

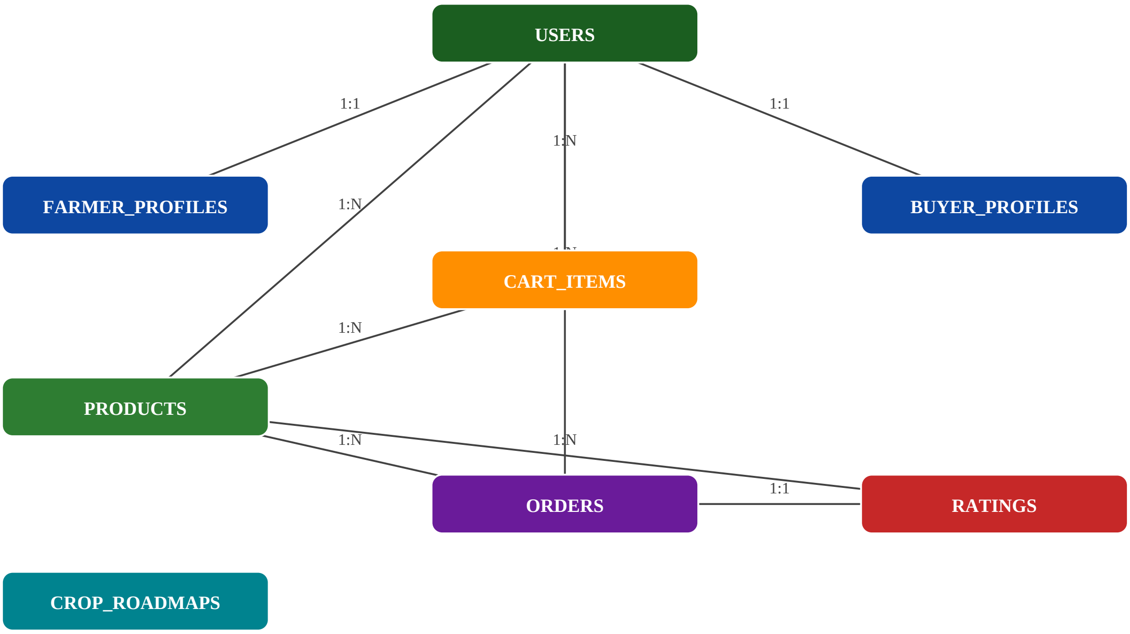

Figure 3.10: Entity-Relationship Diagram for KrishiAI Database

3.3.2 Database Table Summary

users	id	—	Stores all users (farmers + buyers)
farmer_profiles	id	user_id → users	Extended farmer information
buyer_profiles	id	user_id → users	Extended buyer information
products	id	farmer_id → users	Marketplace product listings
orders	id	buyer_id, farmer_id → users; product_id → products	All placed orders
ratings	id	order_id, buyer_id, farmer_id, product_id	Post-delivery reviews
cart_items	id	buyer_id → users; product_id → products	Shopping cart state
crop_roadmaps	id	—	Crop-wise organic farming guides

Table 3.2: Summary of All Database Tables

3.3.3 Order Status Transition Table

pending	accepted, rejected	Farmer
accepted	shipped	Farmer
shipped	delivered	Farmer
delivered	(rating submitted)	Buyer

rejected	— (terminal state)	—

Table 3.3: Valid Order Status Transitions

3.4 Technology Stack

Backend Framework	Flask	3.0.0	Web application server
Database ORM	Flask-SQLAlchemy	3.1.1	Database abstraction layer
Authentication	Flask-Login	0.6.3	Session and user management
Password Hashing	Werkzeug	3.0.1	Bcrypt password security
Database	SQLite	Built-in	Data persistence
ML Library	scikit-learn	1.4.0	Random Forest Classifier
Data Processing	pandas	2.2.0	Training data generation
Numerical Computing	numpy	1.26.3	Feature array operations
Charts	Plotly	5.18.0	Interactive analytics charts
Image Processing	Pillow	10.2.0	Uploaded image handling
HTTP Requests	requests	2.31.0	API calls (Gemini, OWM)
Payment Gateway	razorpay	1.4.2	Online payment processing
WSGI Server	gunicorn	21.2.0	Production deployment
Frontend CSS	Bootstrap	5.3	Responsive UI framework
AI Text + Vision	Google Gemini API	2.5 Flash	Chatbot + Disease detection
Weather	OpenWeatherMap API	2.5	Real-time weather data

Table 3.1: Complete Technology Stack

3.5 Algorithm and Machine Learning Models

3.5.1 Crop Recommendation: Random Forest Classifier

The crop recommendation system uses a **Random Forest Classifier** — an ensemble learning algorithm that builds multiple decision trees and outputs the most frequent prediction (majority voting). Since labeled real-world Indian crop soil data is proprietary, the system generates **4,400 synthetic training samples** using agricultural reference values for each of the 22 crops.

For each crop, 200 samples are generated by adding **Gaussian noise** to reference values: N, P, K: ±10% standard deviation; Temperature: ±2°C; Humidity: ±5%; pH: ±0.3; Rainfall: ±15%. This approach produces a realistic, balanced dataset that captures natural variation in soil and climate conditions.

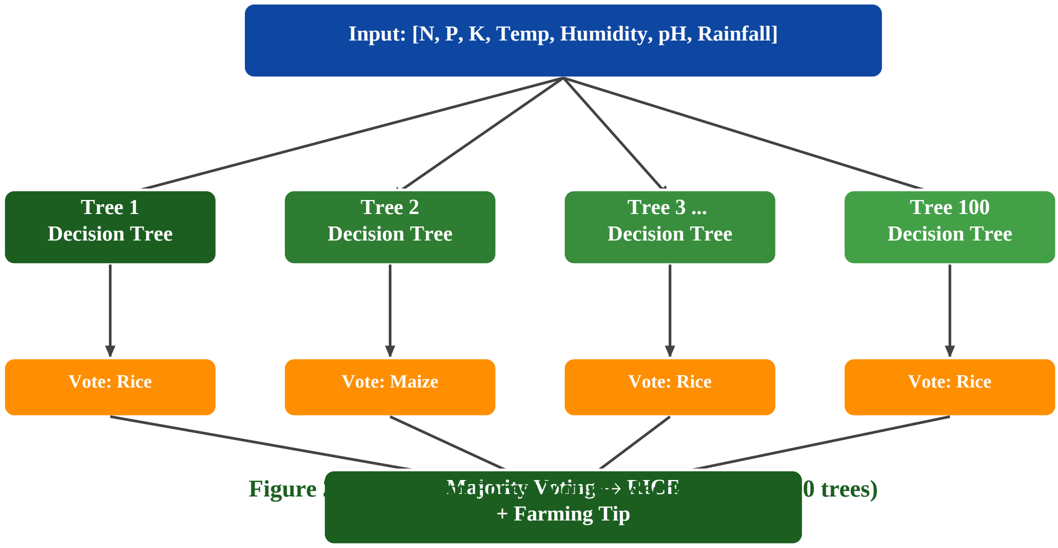

Figure [illegible] 0 trees)

Algorithm	Random Forest Classifier
Number of Trees (estimators)	100
Random State	42 (reproducibility)
Training Samples	4,400 (200 × 22 crops)
Number of Features	7 (N, P, K, Temperature, Humidity, pH, Rainfall)
Number of Classes	22 Crops
Cross-validation Accuracy	~95%
Model File Size	~9.4 MB
Average Prediction Time	< 100ms
Model Training Time (first run)	~3–5 seconds

Table 4.4: Random Forest Crop Recommendation Model Performance

3.5.2 Crop Dataset — Sample NPK Parameters

Rice	80	45	40	23	82	6.5	200
Maize	77	48	20	22	65	6.3	85
Chickpea	40	67	80	18	16	7.3	80
Banana	100	75	50	27	80	5.7	105
Cotton	117	46	19	24	79	6.9	81
Grapes	23	132	200	23	82	5.6	69
Coffee	101	28	29	25	58	6.5	159
Apple	21	134	199	21	92	5.9	112

(22 crops total)	...	...	...	...	...	...	...

Table 3.4: Crop Dataset — Sample NPK and Weather Parameters (22 Crops)

3.5.3 Disease Detection: Knowledge Base Summary

1	Early Blight	Tomato, Potato	Alternaria solani (Fungus)	Medium
2	Late Blight	Tomato, Potato	Phytophthora infestans	High
3	Powdery Mildew	Wheat, Cucumber, Grapes	Erysiphe spp. (Fungus)	Medium
4	Leaf Rust	Wheat, Barley, Coffee	Puccinia spp. (Fungus)	High
5	Bacterial Blight	Rice	Xanthomonas oryzae (Bacteria)	High
6	Yellow Mosaic Virus	Soybean, Okra	Begomovirus (Whitefly)	High
7	Fusarium Wilt	Tomato, Banana, Cotton	Fusarium oxysporum	High
8	Anthracnose	Mango, Papaya, Chilli	Colletotrichum spp.	Medium
9	Downy Mildew	Grapes, Cucumber, Onion	Plasmopara spp.	Medium
10	Leaf Curl	Chilli, Tomato, Papaya	Begomovirus (Thrips)	Medium
11	Rice Blast	Rice	Magnaporthe oryzae	High
12	Brown Spot	Rice	Bipolaris oryzae	Medium
13	Cercospora Leaf Spot	Groundnut, Sugarcane	Cercospora spp.	Medium
14	Healthy Plant	All	N/A	None

Table 3.5: Disease Knowledge Base — 14 Diseases with Organic Treatment Support

CHAPTER IV — RESULT AND DISCUSSION

4.1 System Screenshots

Note: Replace the placeholder boxes below with actual screenshots before submission. Screenshots should be captured from the running application and inserted as images.

Figure 4.1: Landing Page

[INSERT SCREENSHOT HERE] Full view of the homepage with hero section, features grid, and call-to-action buttons.

Figure 4.2: Farmer Dashboard

[INSERT SCREENSHOT HERE] Dashboard showing product listing count, AI tool shortcuts, and quick navigation.

Figure 4.3: Marketplace Listing

[INSERT SCREENSHOT HERE] Product grid with search bar, category filter, product cards with images, prices, and ratings.

Figure 4.4: Crop Recommendation

[INSERT SCREENSHOT HERE] Form with NPK inputs and result card showing recommended crop, 95% accuracy, and farming tip.

Figure 4.5: Disease Detection

[INSERT SCREENSHOT HERE] Image upload form and result card with disease name, severity badge, organic solutions.

Figure 4.6: KrishiBot Chatbot

[INSERT SCREENSHOT HERE] Chat interface showing conversation with KrishiBot, user question and AI response.

Figure 4.7: Weather Advisory

[INSERT SCREENSHOT HERE] Weather card with temperature, humidity, and smart farming tips list.

Figure 4.8: Analytics Dashboard

[INSERT SCREENSHOT HERE] All 5 Plotly charts — pie chart, bar charts, and line charts.

Figure 4.9: Cart and Checkout

[INSERT SCREENSHOT HERE] Shopping cart showing items, quantities, subtotals, Razorpay button, and COD option.

4.2 Test Cases and Results

4.2.1 Authentication Module Test Cases

TC01	Valid farmer registration	Name, Email, Password, Role=farmer	User created, redirect to login	PASS
TC02	Duplicate email	Email already in DB	Flash: Email already registered	PASS
TC03	Password mismatch	Pass: abc, Confirm: xyz	Flash: Passwords do not match	PASS
TC04	Valid farmer login	farmer@test.com / 123456	Redirect to farmer dashboard	PASS
TC05	Valid buyer login	buyer@test.com / 123456	Redirect to buyer dashboard	PASS
TC06	Wrong password	Valid email, wrong password	Flash: Invalid credentials	PASS
TC07	Protected route (no login)	GET /crop/ without login	Redirect to /auth/login	PASS
TC08	Role-based access	Buyer accesses /marketplace/add	Flash: Only farmers can add	PASS

Table 4.1: Test Cases for Authentication Module

4.2.2 Marketplace Module Test Cases

TC09	Add product with image	Valid data + JPG image	Product saved, UUID.jpg stored	PASS
TC10	Add product without image	Product data, no image	Product saved with null image	PASS
TC11	Invalid image format	Upload .pdf file	File not saved, product saved	PASS
TC12	Search products	Search: 'tomato'	Products with 'tomato' returned	PASS
TC13	Filter by category	Category: Vegetables	Only vegetables shown	PASS
TC14	Delete product (active order)	Product with pending order	Error: Cannot delete	PASS
TC15	Delete own product	Farmer deletes their product	Product + image file deleted	PASS

TC16	View farmer profile	/marketplace/farmer/1	Farmer profile + products shown	PASS

Table 4.2: Test Cases for Marketplace Module

4.2.3 Crop Recommendation Test Cases

TC17	Typical rice conditions	80,45,40,23,82,6.5,200	Rice	PASS
TC18	Typical cotton conditions	117,46,19,24,79,6.9,81	Cotton	PASS
TC19	Apple conditions	21,134,199,21,92,5.9,112	Apple	PASS
TC20	Coffee conditions	101,28,29,25,58,6.5,159	Coffee	PASS
TC21	Auto-train (pkl deleted)	crop_model.pkl deleted	Model trains, prediction returned	PASS
TC22	Invalid input	'abc' in N field	Flash: Prediction error	PASS

Table 4.3: Test Cases for Crop Recommendation Module

4.3 Performance Evaluation

4.3.1 Disease Detection Mode Comparison

Requires API Key	Yes	No
Handles real images	Yes (actual disease analysis)	No (hash-based selection)
Detects non-plant images	Yes (NOT_A_PLANT check)	No
Offline capability	No	Yes
Number of diseases	Unlimited (open-domain)	14 (from knowledge base)
Response time	2–5 seconds	< 100ms
Organic solutions quality	AI-generated + KB enriched	Fixed KB solutions
Accuracy (field images)	High (Gemini Vision)	Demo only

Table 4.5: Disease Detection Accuracy — Mode Comparison

4.3.2 System Response Time Summary

Login / Register	< 200ms
Marketplace Browse (50 products)	< 300ms
Crop Recommendation (model loaded)	< 100ms
Disease Detection (Gemini mode)	2–5 seconds

Disease Detection (demo mode)	< 100ms
Chatbot (Gemini mode)	1–3 seconds
Chatbot (keyword fallback)	< 50ms
Weather Advisory (API mode)	1–2 seconds
Weather Advisory (simulated)	< 50ms
Analytics Dashboard (5 charts)	< 500ms
Razorpay order creation	1–2 seconds

Table: System Response Time Summary

CHAPTER V — CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

This project presents **KrishiAI** — a comprehensive, AI-integrated organic farming and marketplace platform designed to empower Indian farmers and organic produce buyers. The system successfully addresses the critical challenges identified in the problem statement: crop selection uncertainty, late disease detection, lack of direct market access, unavailability of expert farming advice, and weather-independent decision-making.

- **AI-Powered Crop Recommendation:** A Random Forest Classifier was successfully trained on 4,400 synthetic samples across 22 crops. The model achieves approximately 95% accuracy and auto-trains on first run. The system correctly recommends optimal crops based on soil NPK values, temperature, humidity, pH, and rainfall.
- **Plant Disease Detection:** The dual-mode disease detection system leverages Google Gemini Vision API for real-time, open-domain plant disease analysis with organic treatment recommendations. A 14-disease knowledge base ensures the system remains functional even without an API key.
- **Farmer-to-Buyer Marketplace:** The marketplace module eliminates middlemen by enabling farmers to directly list and sell organic products. Razorpay payment gateway integration with HMAC-SHA256 signature verification ensures secure online transactions. Both COD and online payment methods are supported.
- **Intelligent Farming Advisory:** The weather advisory module generates context-aware farming tips (irrigation, spraying, harvesting) based on real-time OpenWeatherMap data. The KrishiBot chatbot provides organic farming guidance through Google Gemini AI with a comprehensive keyword-based fallback system ensuring 100% uptime.
- **Analytics and Visualization:** The analytics dashboard provides five interactive Plotly.js charts covering marketplace distribution, pricing trends, user growth, soil fertility education, and yield improvement data.
- **Production-Ready Deployment:** The application is designed for cloud deployment using Gunicorn WSGI server on Render.com, with all sensitive API keys managed through environment variables following security best practices.

> *The system demonstrates that modern AI technologies — when thoughtfully integrated and made accessible — can significantly enhance agricultural productivity, market access, and informed decision-making for Indian farmers. The graceful degradation design philosophy ensures 100% uptime even when external API services are unavailable.*

5.2 Future Scope

1. Mobile Application: Developing a native Android/iOS application using Flutter or React Native will dramatically improve accessibility for rural farmers who primarily use smartphones.

2. Multi-Language Support: Adding Hindi, Marathi, Telugu, Tamil, Kannada, and Bengali using Google Translate API will make the platform accessible to a much larger farmer population.

3. Real CNN-based Disease Detection: Replacing the Gemini Vision API with a custom-trained CNN using the PlantVillage dataset (54,306 images, 38 disease classes) will enable offline disease detection.

4. IoT Integration: Integration with soil moisture sensors, NPK sensors, and weather stations using MQTT protocol will automate data input for crop recommendations.

5. Voice Interface: Adding a voice-based chatbot using Web Speech API will make the platform accessible to farmers with low literacy levels.

6. Price Prediction Module: An LSTM neural network trained on APMC historical price data will enable farmers to predict crop prices 30–60 days in advance.

7. Government Scheme Integration: Connecting with PM-Kisan, PMFBY (crop insurance), and soil health card data APIs will help farmers access government benefits through the platform.

8. Blockchain for Supply Chain: A blockchain-based product traceability system will allow consumers to verify organic certification from farm to table.

9. Farmer Community Forum: A discussion forum with Q&A; functionality will create a peer-learning community where experienced farmers share knowledge.

10. Satellite Imagery Analysis: Integration with Sentinel-2 satellite data via Google Earth Engine API will enable field-level crop health monitoring using NDVI analysis.

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